STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define pressure.
2. SI unit of dynamic viscosity?
3. Continuity relation for incompressible flow?
4. Bernoulli relation?
5. Surface tension unit?
6. Pressure rise at depth h?
7. Stokes drag?
8. Excess pressure in a liquid drop?

Section B 2-mark questions - 6 x 2 = 12 marks

9. Water at 3m depth, rho=1000,g=10: gauge pressure?
10. If pipe area halves, incompressible speed?
11. Under Bernoulli assumptions, faster horizontal flow means pressure?
12. Force from p=5000 Pa over area .02 m^2?
13. Drop S=.06 N/m,r=.003 m: excess pressure?
14. Why is a soap bubble pressure difference twice a drop's?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Pipe area 4 cm^2 at 2 m/s narrows to 1 cm^2. New speed?
16. Horizontal water speeds 2 and 6 m/s, rho=1000. Find p1-p2.
17. Stokes drag for r=.5 mm,eta=1 Pa s,v=.02 m/s?
18. Explain why pressure rises with depth.

Section D 4-mark questions - 2 x 4 = 8 marks

19. Areas 6 cm^2 and 2 cm^2; inlet speed 1.5 m/s. Find outlet speed.
20. Soap bubble r=2mm,S=.03N/m: excess pressure?